

Project Design Phase

Solution Architecture

Date	02 July 2026
Team ID	SWTID-2026-4978
Project Name	Strategic Product Placement Analysis
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Proposed Visual BI Solution Architecture Description:

To replace the non-applicable cloud voice application example, our business intelligence solution utilizes a structured 3-tier local-to-cloud analytical pipeline designed for high-performance data processing:

1. Data Ingestion Tier (Data Source)

The architecture begins with the storage of the unorganized transactional log records (Product Positioning.csv). This serves as the baseline data repository layer containing spatial coordinates, sales volumes, pricing fields, and consumer categories.

2. Analytical & Calculation Tier (Processing Engine)

The local workspace layer runs on the high-speed **Tableau VizQL & Hyper Extract Engine**. This component isolates multi-variable dimensions, computes calculated parameters (such as promotion volume lift vs. baseline shelf location lift), and maps store geometries out across interconnected relational tables.

3. Visualization & Presentation Tier (User Delivery Interface)

The final tier outputs the processed metrics into a single, cohesive 8-sheet interactive story layout. Category planners and store operations managers interact directly with web containers via global cross-sheet dashboard filters. This layer is published to **Tableau Cloud / Server**, allowing stakeholders to safely drill down into demographic segments (Families, Students, Seniors) and download clean summaries.